

SECONDARY SCIENCE EXAMINATION- 2015

(ANNUAL)

SCIENCE

[TIME: 2 HOURS 30 minutes]

F. M. 80

Instructions for Candidates:

- (1) Candidates are required to give their answers in their own words as far as possible.
- (2) Figures in the right hand margin indicate full marks.
- (3) While answering the questions Candidate should adhere to the word limit as far as possible.
- (4) 15 minutes of extra time has been allotted for the candidates to read the questions carefully.
- (5) This question paper is divided into two sections section A and section B.

SECTION – A

1. What is snell's law?

[1]

Ans: Snell's law states that the ratio of the sines of the angles of incidence and refraction is equivalent to the ratio of phase velocities in the two media, or equivalent to the reciprocal of the ratio of the indices of refraction.

2. An ammeter is added in which order?

[1]

Ans: An ammeter is connected in series with the circuit because the purpose of the ammeter is to measure the current through the circuit.

3. Which lens is called converging lens?

[1]

Ans: A convex lens is called a converging lens because when parallel rays of light fall on such a lens, the rays will get refracted and after refraction they will converge.

4. What type of electric current is used in electric heater?

[1]

Ans: Direct current is used in electric heater.

5. In which mirror only imaginary reflection is formed?

[1]

Ans: In Plane mirrors imaginary reflection is formed.

6. Write two uses of convex concave mirror. [2]

Ans: The concave mirror is a converging mirror, so that it is used for many purposes , It is used as a torch to reflect the light, It is used in the aircraft landing at the airports to guide the airplanes.

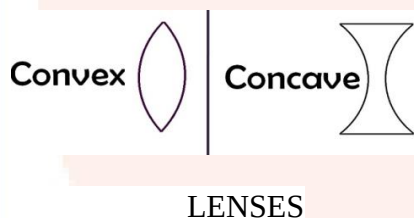
7. What is collective action of resistor? Describe the electric circuit. [2]

Ans: A closed and continuous path through which electric current flows is known as Electrical circuits.

8. Explain the difference in the convex lens and concave lens with diagram. [3]

Ans: A lens, which is thicker at the center and thinner at its end is called convex lens. A convex lens is also known as converging lens because it converges a beam of light rays passing through it.

A lens which is thinner at the center and thicker at its end is called concave lens. A concave lens is known as diverging lens because its diverges a parallel beam of light rays passing through it

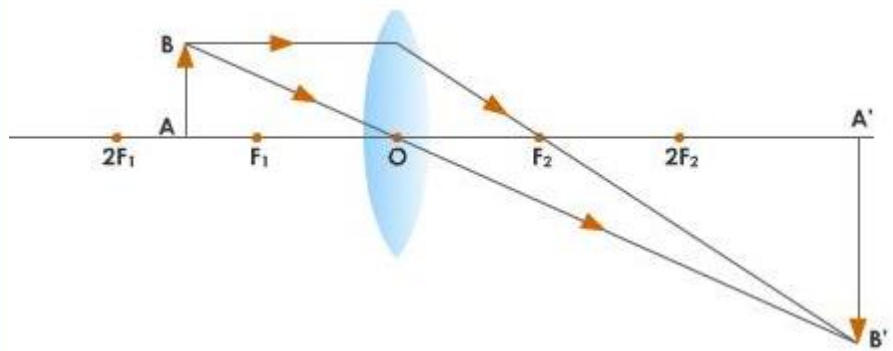


9. Why passive gas is filled in electric bulb? [3]

Ans: The filament of electric bulb is filled with chemically inactive nitrogen and Argon gas to prolong the life of filament.

10. Prove $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$ in convex lens.

Ans:



LENS FORMULA

Let AB represent an object placed at right angles to the principal axis at a distance greater than the focal length f of the convex lens. The image A^1B^1 is formed beyond $2F_2$ and is real and inverted.

OA = Object distance = u

OA^1 = Image distance = v

OF_2 = Focal length = f

$\triangle OAB$ and $\triangle OA^1B^1$ are similar

$\therefore \angle BAO = \angle B^1A^1O = 90^\circ, \angle AOB = \angle A^1OB^1$ (Vertically opposite angles)

$\therefore \angle ABO = \angle A^1B^1O$

$$\therefore \frac{A^1B^1}{AB} = \frac{OA^1}{OA} \dots\dots\dots(1)$$

Similarly $\triangle OCF_2$ and $\triangle F_2A^1B^1$ are similar

$$\therefore \frac{A^1B^1}{OC} = \frac{F_2A^1}{OF_2}$$

But we know that $OC = AB$

$\therefore$ The above equation can be written as

$$\therefore \frac{A^1B^1}{OC} = \frac{A^1B^1}{AB} = \frac{F_2A^1}{OF_2}$$

$$\Rightarrow \frac{A^1B^1}{AB} = \frac{F_2A^1}{OF_2} \dots\dots\dots(2)$$

From equation (1) and (2), we get

$$\frac{OA^1}{OA} = \frac{F_2 A^1}{OF_2} = \frac{OA^1 - OF_2}{OF_2}$$

$$\frac{v}{-u} = \frac{v - f}{f}$$

$$vf = -u(v - f)$$

$$vf = -uv + uf \dots\dots\dots(3)$$

Dividing equation (3) throughout by uvf

$$\frac{1}{u} = -\frac{1}{f} + \frac{1}{v}$$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

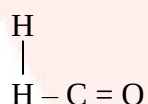
11. Write the molecular formula of plaster of paris.

Ans: $CaSO_4 \cdot 1/2 H_2O$.

12. Writing the names of two metals that are found in Free State in nature.

Ans: Gold and silver are the two metals that are found in Free State in nature

13. Write IUPAC Name of following-



Ans: Formaldehyde.

14. Which acid is found in the ant's sting?

Ans: Formic acid is found in the ant's sting.

15. How many vertical columns are found in the periodic table?

Ans: The vertical columns on the periodic table are called groups. There are 18 groups on the periodic table,

16. Why are noble gases placed in different job?

Ans: The noble gases are placed in a separate group in the periodic table because they are extremely non-reactive. They can occur in nature by themselves and don't need another element to stabilize them. Thus they fit into a different category to the other non-metals.

17. Why is conversion of Ethanol to ethanoic acid called an Oxidation reaction?

Ans: H₂ molecule is removed from ethanol to form ethanal. As loss of Hydrogen is oxidation so, the reaction is an oxidation reaction. Similarly, in second step an Oxygen atom is added to form ethanoic acid from ethanal. As, gain of Oxygen is called oxidation so, the reaction is an oxidation reaction.

18. Write the method of methane gas formation in the laboratory and its chemical reaction with chlorine.

Ans: By the hydrolysis of aluminum carbide:

In laboratory methane can be prepared by boiling aluminum carbide with water.



19. Platinum, gold and silver are used in making Jewellery. Why?

Ans: Gold is used to make Jewellery because of the attractiveness of its luster and because of its rarity. It also doesn't tarnish and doesn't react with air. Gold is malleable. It can be beaten into very thin sheets. Platinum is a naturally white precious metal and has several properties that make it a preferred jewellery material. The main advantage of using platinum for jewellery are its strength and resistant to tarnish.

Produce ethanol by fermentation method. Write two uses of Ethanol. 5Ans: Ethanol is obtained by the fermentation of molasses which are obtained from sugarcane juice.

Uses of ethanol are-

- Ethanol is used as an active ingredient in all alcoholic drinks.
- It is useful in medicines like tincture of Iodine cough syrups

OR

Mention the principle of the Zinc extraction from Zinc.

Ans: The most commonly available for used in the extraction of zinc is called zinc blend ore zinc sulphide. The best concentration method of zinc Ore is known as froth flotation presentation of Froth floatation procedure is mainly used to remove gauge from sulphide ore. Ore is powdered and a suspension is created in water. The main ingredients of the fourth flotation are collectors and stabilizers. The frothy metal is removed from the top and dried to recover the metals. The next step in the extraction of zinc is called roasting. The concentrated Ore is finely ground into

small pieces and then suspended in a rising steam in air. This converts the zinc blend to Zinc oxide which is more reactive. Zinc oxide is a mix with coke and ground to fine pieces. And heated in a process which is known as Belgian process. Zinc oxide gets reduced to metallic zinc.

21. Which tissue support in water plants in plants? 1

Ans: Xylem which tissue support in water transport in plants?

22. What is the creative and functional unit of renal? 1

Ans: Nephrons are the functional unit of renal.

23. Where is the powerhouse of the cell? 1

Ans: Mitochondria is the powerhouse of the cell.

24. What do we call the respiratory action in the absence of oxygen? 1

Ans: Anaerobic respiration occurs in the absence of oxygen.

25. When did the Ganga cleaning scheme begin? 1

Ans: The clean Ganga Project was started with a significant amount of fanfare in 2014 after Narendra Modi became the Prime Minister of India.

26. What disease is caused due to iodine deficiency and how? 2

Ans: Iodine in our diet is essential for or secretion of thyroxine hormone .In case the diet iodine deficient there occurs a disease called goitre which is characterized by swelling neck.

27. How Terrestrial organism and aquatic organism get oxygen for respiration? 2

Ans: Aquatic animals utilize oxygen dissolved in water for respiration .The aquatic organism breath more rapidly to accumulate more and more oxygen. Fishes take in water through diet through their mouth and force it past the gills where the dissolved oxygen is taken by the blood.

Terrestrial organism use atmospheric oxygen for respiration which is absorbed by different organs in different on animals. All these organs have a structure that increases surface area and is in the contact with oxygen rich atmosphere.

28. Write the role of components in the ecosystem? 2

Ans: An ecosystem is made up of two main components.

Abiotic components - Non- living features of ecosystem on which an organism depends are abiotic components which include physical factors like light, temperature, wind etc.

Biotic components - It includes the entire living organism present in the ecosystem it includes mainly three categories of organisms.

Producers make their own food for example plants.

Consumers depend on producers for their nutritional requirement for example grazing animals, human.

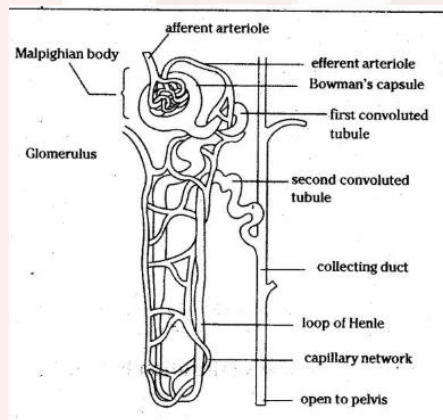
Decomposers are organisms of decay. They feed on decaying and dead matter for example: bacteria, fungi.

29. What is excretion? Describe two major parts of it.

Ans: Excretion is the biological process by which an organism removes harmful metabolic waste from the body. The mode of excretion is completely different in unicellular and multicellular organisms. The main organ of human excretory system is kidney.

30. Describe the different parts of human nephron with clear diagram.

Ans: Each kidney is made up of thousands of tiny tubules called nephron. Nephron is the structural and functional unit of kidney. Blood at high pressure travels into these tubules by blood capillaries called glomerulus which are surrounded by Cup shaped capsule call Bowman's capsule.



NEPHRON

Section B

i) If 120 coulomb charge is flowing in 1 minute from any bulb, then the value of the current is :-

- (a) 1 ampere (b) 2 ampere (c) 3 ampere (d) 4 ampere

Ans: - 2 Ampere

ii) What is used to get the Spectrum?

(a) Slide of glass (b) Concave Mirror (c) Convex Lens (d) Prism

Ans: - Prism

iii) What is the S.I Unit of Resistance?

(a)Joule (b) Volt (c) Ohm (d) Ampere

Ans: - Ohm

(iv) The electric current in the circuit is measured by:-

(a)Voltmeter (b) Ammeter (c) Galvanometer (d) none of these

Ans: - Ammeter

(v)Which type of mirror is used for shaving?

(a) Plane mirror (b) Concave mirror (c) Convex mirror (d) None of these

Ans: - Concave mirror

(vi) Which device does not have a sign of positive and negative?

(a) In ammeter (b) In voltmeter (c) In the loop (d) In the electric cell

Ans: - In the loop

(vii)Which is used in Simple Microscope?

(a)Concave mirror (b) Convex mirror (c) Concave lens (d) Convex lens

Ans: - Convex lens

(viii) What is the chemical formula of Blue Vitriol?

(a) $\text{CuSO}_4 \cdot 7\text{H}_2\text{O}$ (b) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (c) $\text{CuSO}_4 \cdot 4\text{H}_2\text{O}$ (d) $\text{CuSO}_4 \cdot 10\text{H}_2\text{O}$

Ans: - $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$

(ix) Which of the following is the salt?

(a)HCL (b) NaOH (c) K_2SO_4 (d) NH_4OH

Ans: - K_2SO_4

(x) How many covalent bonds are there in Nitrogen molecule?

- (a) 1 (b) 2 (c) 4 (d) 3

Ans:-3

(xi) What is the chemical formula of Marble?

- (a) CaCO_3 (b) $\text{Mg}(\text{HCO}_3)$ (c) $\text{Ca}(\text{HCO}_3)_2$ (d) $\text{Mg}(\text{CO}_3)$

Ans: - CaCO_3

(xii)_____ is an example of forest estate.

- (a) Soil (b) Wood (c) Copper (d) Aluminium

Ans: - Wood

(xiii)What is the chemical formula of Sugar?

- (a) CH_3COOH (b) $\text{C}_6\text{H}_{12}\text{O}_6$ (c) $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ (d) CH_3CHO

Ans: - $\text{C}_{12}\text{H}_{22}\text{O}_{11}$

(xiv) When solution of Methyl Orange is added to the aqueous solution of Sodium carbonate, then the colour of solution converted into which colour?

- (a) Yellow (b) Red (c) Green (d) Blue

Ans: Yellow

(xv) Which is essential for Autotrophic Nutrition?

- (a) CO_2 (b) Chlorophyll (c) Sunlight (d) None of these

Ans: - Sunlight

(xvi) Lack of Insulin causes:-

- (a) Goiter (b) Dwarfism (c) Diabetes (d) None of these

Ans: - Diabetes

(xvii) In which the open circulatory system is found?

- (a) In human (b) In Cockroach (c) In horse (d) In camel

Ans: - In cockroach

(xviii) Euro II is related to:-

(a) Air pollution (b) Water pollution (c) Soil pollution (d) none of these

Ans: - None of these

(xix) Which of the following is non-biodegradable substance:-

(a) DDT (b) Paper (c) washed stool (d) none of these

Ans: - DDT

(xx) The colour of Chlorophyll pigment is:-

(a) Green (b) Blue (c) Red (d) White

Ans: - Green